TENSOR-COMPATIBILITY CHARACTERIZES QUASI-COHERENT MODULES ON THE AFFINE \'ETALE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites two Descent lemmas, one Modules on Sites lemma and one Schemes lemma, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Faithful flatness of standard covering $\check{\mathrm{C}}$ech algebra. Stacks Project, Descent, Lemma descent-lemma-standard-covering-Cech. Shows $\mathcal{O}_X(U) \to \prod \mathcal{O}_X(U_i)$ is faithfully flat.
- Exactness after faithful flat base change. Stacks Project, Descent, Lemma descent-lemma-ff-exact. Gives exactness of the tensor-amplified sequence.
- Intrinsicness of quasi-coherent modules on ringed sites. Stacks Project, Modules on Sites, Lemma sites-modules-lemma-special-locally-free. Supports the (2)-(3) transfer along the site equivalence.
- Tensor of sections with pullback. Stacks Project, Schemes, Lemma schemes-lemma-widetilde-pullback. Concludes (1) from (3).

Source reference. Original reference 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original equation 0H03: sources/stacks-chow/tags.txt, source lines 21198–21198. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03FI: sources/stacks-chow/tags.txt, source lines 4493–4493. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 023M: sources/stacks-chow/tags.txt, source lines 2747–2747. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DM: sources/stacks-chow/tags.txt, source lines 4425–4425. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01I9: sources/stacks-chow/tags.txt, source lines 1970–1970. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $\mathcal{F}$ be a presheaf of $\mathcal{O}_X$-modules on $X_{affine,étale}$. The following are equivalent*

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(1) *for every morphism* $U \to U'$ *of* $X_{affine,étale}$ *the map* $\mathcal{F}(U') \otimes_{\mathcal{O}_X(U')} \mathcal{O}_X(U) \to \mathcal{F}(U)$ *is an isomorphism,*
(2) $\mathcal{F}$ *is a quasi-coherent module on the ringed site* $(X_{affine,étale}, \mathcal{O}_X)$ *in the sense of Modules on Sites, Definition 03DL,*
(3) $\mathcal{F}$ *corresponds to a quasi-coherent module on* X *via the equivalence (0H03),*

Proof. Assume (1) holds. To show that $\mathcal{F}$ is a sheaf, let $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ be a covering of $X_{affine,étale}$. The sheaf condition for $\mathcal{F}$ and $\mathcal{U}$, by our assumption on $\mathcal{F}$, reduces to showing that

$$0 \to \mathcal{F}(U) \to \prod \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{O}_X(U_i) \to \prod \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{O}_X(U_i \times_U U_j)$$

is exact. This is true because $\mathcal{O}_X(U) \to \prod \mathcal{O}_X(U_i)$ is faithfully flat (by Descent, Lemma 03FI and the fact that coverings in $X_{affine,étale}$ are standard étale coverings) and we may apply Descent, Lemma 023M. Next, we show that $\mathcal{F}$ is quasi-coherent on $X_{affine,étale}$. Namely, for U in $X_{affine,étale}$, set $R = \mathcal{O}_X(U)$ and choose a presentation

$$\bigoplus\nolimits_{k\in K} R \longrightarrow \bigoplus\nolimits_{l\in L} R \longrightarrow \mathcal{F}(U) \longrightarrow 0$$

by free R-modules. By property (1) and the right exactness of tensor product we see that for every morphism $U' \to U$ in $X_{affine,étale}$ we obtain a presentation

$$\bigoplus\nolimits_{k\in K} \mathcal{O}_X(U') \longrightarrow \bigoplus\nolimits_{l\in L} \mathcal{O}_X(U') \longrightarrow \mathcal{F}(U') \longrightarrow 0$$

In other words, we see that the restriction of $\mathcal{F}$ to the localized category $X_{affine,etale}/U$ has a presentation

$$\bigoplus\nolimits_{k\in K} \mathcal{O}_X|_{X_{affine,étale}/U} \longrightarrow \bigoplus\nolimits_{l\in L} \mathcal{O}_X|_{X_{affine,étale}/U} \longrightarrow \mathcal{F}|_{X_{affine,étale}/U} \longrightarrow 0$$

as required to show that $\mathcal{F}$ is quasi-coherent. With apologies for the horrible notation, this finishes the proof that (1) implies (2).

Since the notion of a quasi-coherent module is intrinsic (Modules on Sites, Lemma 03DM) we see that the equivalence (0H03) induces an equivalence between categories of quasi-coherent modules. Thus we have the equivalence of (2) and (3).

Let us assume (3) and prove (1). Namely, let $\mathcal{G}$ be a quasi-coherent module on X corresponding to $\mathcal{F}$. Let $h : U \to U' \to X$ be a morphism of $X_{affine,étale}$. Denote $f : U \to X$ and $f' : U' \to X$ the structure morphisms, so that $f = f' \circ h$. We have $\mathcal{F}(U') = \Gamma(U', (f')^*\mathcal{G})$ and $\mathcal{F}(U) = \Gamma(U, f^*\mathcal{G}) = \Gamma(U, h^*(f')^*\mathcal{G})$. Hence (1) holds by Schemes, Lemma 01I9. □